实验 2：数据定义与数据操纵（实验报告）

（2026 春季，智能软件与工程学院）

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（说明：实验报告请使用 PDF 格式提交，截止日期：2026.05.12）

实验环境

[如果你不是使用本课程建议的 MySQL 数据库，请用一句话介绍你使用的操作系统、数据库管理系统、软件版本]

实验过程

[所有要求提交的 SQL 语句文本和执行结果截图]

第一部分：数据库创建

```
use mysql;
drop schema if exists myschool26;
create schema myschool26 charset gb2312 collate gb2312_bin;
use myschool26;

create table students(
    sno char(9) not null,
    sname char(20) not null,
    ssex char(2) not null,
    birthday date not null,
    dept char(20),
    enrolldate date,
    primary key (sno),
    check (ssex in ('男', '女'))
);

create table teachers(
    tno char(9) not null,
    tname char(20) not null,
    dept char(20),
    primary key (tno)
);
```

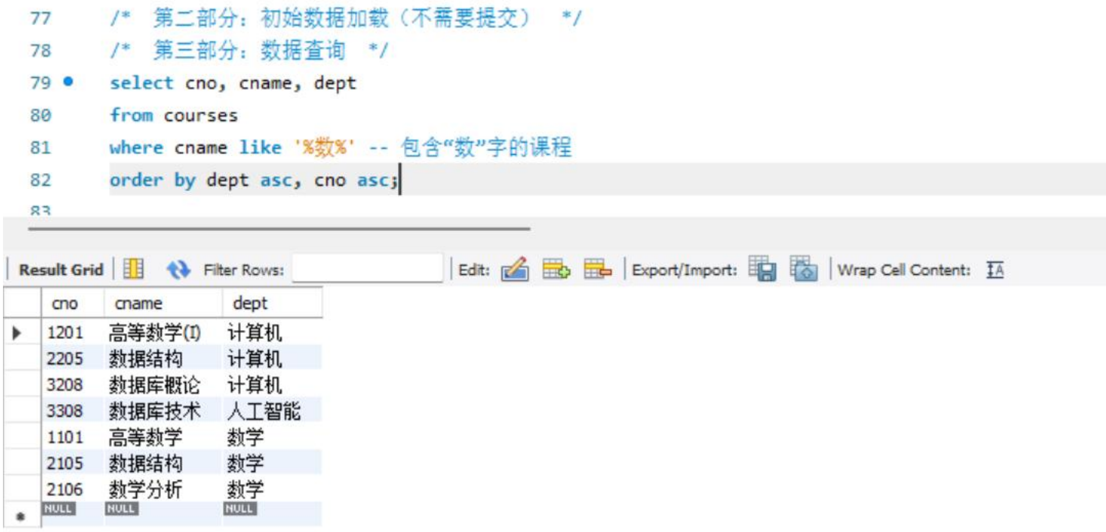
```
create table courses(
	cno char(8) not null,
	cname char(40) not null,
	credit int not null,
	chours int not null,
	dept char(20),
	primary key (cno),
	check (credit > 0 and credit < 10)
);

create table courseclass(
	clsno char(8) not null,
	cno char(8) not null,
	tno char(9),
	clsyear int not null,
	clsterm char(4) not null,
	primary key (clsno),
	check (clsterm in ('春季', '秋季', '暑期')),
	foreign key (cno) references courses(cno)
		on update cascade
		on delete restrict,
	foreign key (tno) references teachers(tno)
		on update cascade
		on delete set null
);

create table enrollment(
	sno char(9) not null,
	clsno char(8) not null,
	grade int,
	primary key (sno, clsno),
	foreign key (sno) references students(sno)
		on update cascade
		on delete restrict,
	foreign key (clsno) references courseclass(clsno)
		on update cascade
		on delete restrict,
	check (grade >= 0 and grade <= 100)
);
```

第二部分：初始数据加载（不需要提交）

第三部分：数据查询

1.

select cno, cname, dept
from courses
where cname like '%数%' -- 包含“数”字的课程
order by dept asc, cno asc;



cno	cname	dept
1201	高等数学(I)	计算机
2205	数据结构	计算机
3208	数据库概论	计算机
3308	数据库技术	人工智能
1101	高等数学	数学
2105	数据结构	数学
2106	数学分析	数学
NULL	NULL	NULL

2.

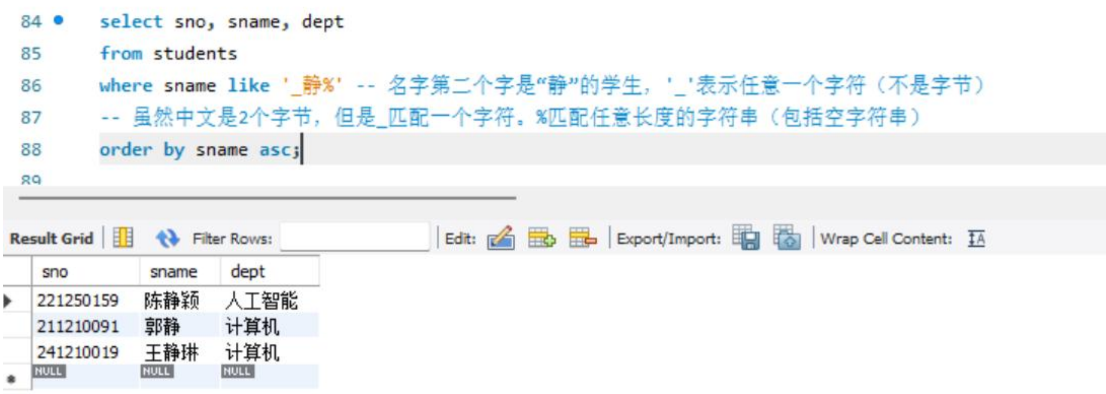
select sno, sname, dept

from students

where sname like '_静%' -- 名字第二个字是“静”的学生，'_'表示任意一个字符（不是字节）

-- 虽然中文是 2 个字节，但是_匹配一个字符。%匹配任意长度的字符串（包括空字符串）

order by sname asc;



sno	sname	dept
221250159	陈静颖	人工智能
211210091	郭静	计算机
241210019	王静琳	计算机
NULL	NULL	NULL

3.

select s.sno, s.sname, c.cno, c.cname

from enrollment e

join students s on e.sno = s.sno -- 必须要 join，因为 sname 这些值在 enrollment 表里没有，必须要连接到 students 表才能获取

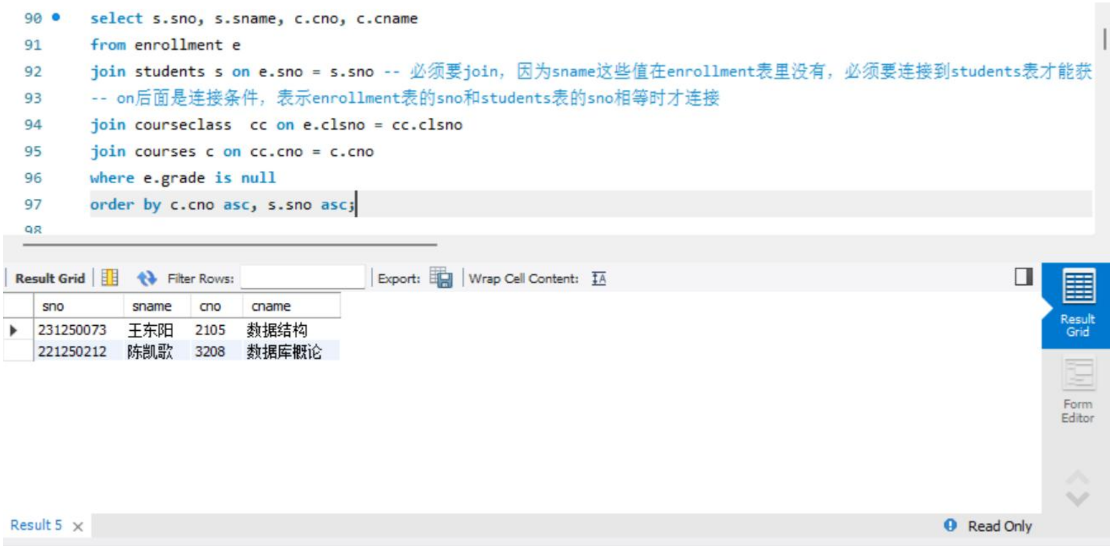
-- on 后面是连接条件，表示 enrollment 表的 sno 和 students 表的 sno 相等时才连接

join courseclass   cc on e.clsno = cc.clsno

join courses c on cc.cno = c.cno

where e.grade is null

order by c.cno asc, s.sno asc;



4.

select s.sno, s.sname,

    count(*) as course_count, -- 统计每个学生选了多少门课程，直接用*就行，因为每一行代表选了一门课了

    sum(c.credit) as total_credits, -- 统计每个学生选了多少学分，sum 函数会自动忽略 null 值，所以不需要在 where 里加 grade is not null 的条件了

    avg(e.grade) as avg_grade

from enrollment e

join students s on e.sno = s.sno

join courseclass   cc on e.clsno = cc.clsno

join courses c on cc.cno = c.cno

where year(s.enrolldate) = 2023

  and e.grade is not null

group by s.sno

having min(e.grade) >= 60 -- 这里不能放到 where 里，因为 where 是在分组前过滤数据的，而 having 是在分组后过滤数据的，只有分组后才能计算每个学生的最低成绩

-- 聚合函数不能直接在 where 里使用，必须要在 group by 后面使用 having 来过滤分组后的结果

order by s.sno asc;

```sql
select s.sno, s.sname,
       count(*) as course_count, -- 统计每个学生选了多少门课程，直接用*就行，因为每一行代表选了一门课了
       sum(c.credit) as total_credits, -- 统计每个学生选了多少学分，sum函数会自动忽略null值，所以不需要在where里
       avg(e.grade) as avg_grade
from enrollment e
join students s on e.sno = s.sno
join courseclass  cc on e.clsno = cc.clsno
join courses c on cc.cno = c.cno
where year(s.enrolldate) = 2023
  and e.grade is not null
group by s.sno
having min(e.grade) >= 60 -- 这里不能放到where里，因为where是在分组前过滤数据的，而having是在分组后过滤数据的，只
-- 聚合函数不能直接在where里使用，必须要在group by后面使用having来过滤分组后的结果
order by s.sno asc;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

sno	sname	course_count	total_credits	avg_grade
231220042	李语诚	3	11	83.0000
231250006	吴晓静	4	17	79.5000
231250013	吴川江	3	11	81.0000
231250020	吴雅婷	3	11	77.3333
231250021	吴迪	3	12	90.6667
231250045	蒋权友	4	17	86.2500
231250054	郑韬	3	11	91.6667
231250064	李依娜	3	12	79.0000
231250073	王东阳	3	13	83.3333

5.

select t.tno, t.tname,

count(*) as courseclass_cnt,

sum(c.chours) as total_hours

from courseclass cc

join teachers t on cc.tno = t.tno

join courses c on cc.cno = c.cno

where cc.tno is not null

group by t.tno

order by total_hours desc, t.tno asc;

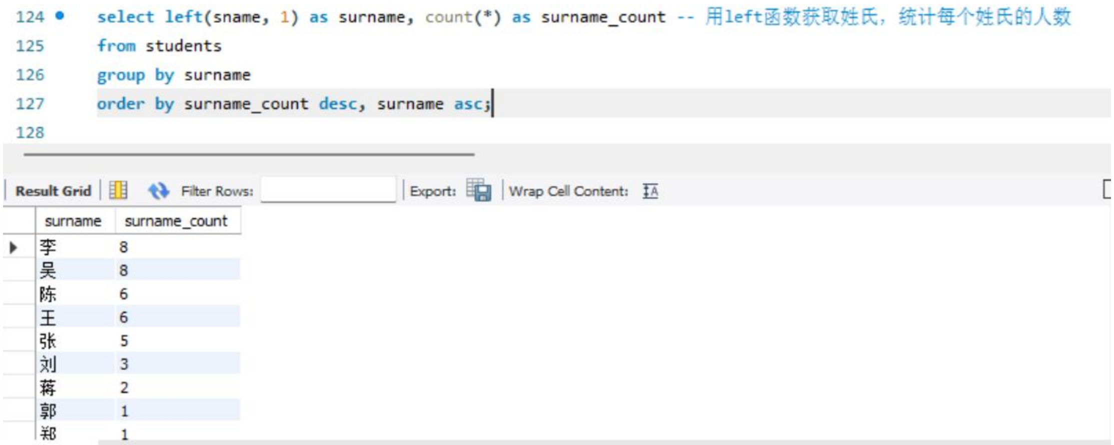
```sql
select t.tno, t.tname,
count(*) as courseclass_cnt,
sum(c.chours) as total_hours
from courseclass cc
join teachers t on cc.tno = t.tno
join courses c on cc.cno = c.cno
where cc.tno is not null
group by t.tno
order by total_hours desc, t.tno asc;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

tno	tname	courseclass_cnt	total_hours
601	王家庚	8	576
602	朱荣华	6	486
802	张无忌	7	432
805	杨玉昆	3	324
704	张家荣	5	236
703	陈坤	4	216
811	李晋城	2	180

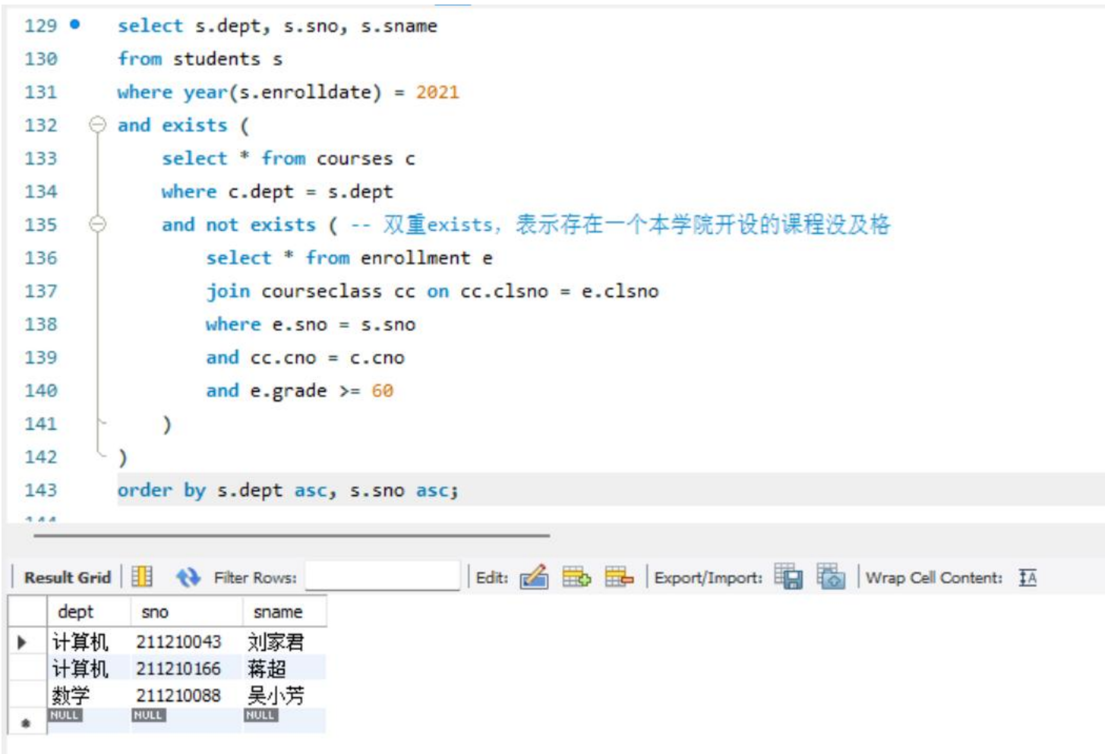
6.

select left(sname, 1) as surname, count(*) as surname_count -- 用 left 函数获取姓氏，统计每个姓氏的人数
from students
group by surname
order by surname_count desc, surname asc;



7.

select s.dept, s.sno, s.sname
from students s
where year(s.enrolldate) = 2021
and exists (
    select * from courses c
    where c.dept = s.dept
    and not exists (-- 双重 exists，表示存在一个本学院开设的课程没及格
        select * from enrollment e
        join courseclass cc on cc.clsno = e.clsno
        where e.sno = s.sno
        and cc.cno = c.cno
        and e.grade >= 60
)
)
order by s.dept asc, s.sno asc;



8.

```
select s.dept, s.sno, s.sname
from students s
where year(s.enrolldate) = 2021
and not exists ( -- 就是加一个 not
	select * from courses c
	where c.dept = s.dept
	and not exists (
		select * from enrollment e
		join courseclass cc on cc.clsno = e.clsno
		where e.sno = s.sno
		and cc.cno = c.cno
		and e.grade >= 60
	)
)
order by s.dept asc, s.sno asc;
```

```sql
select s.dept, s.sno, s.sname
from students s
where year(s.enrolldate) = 2021
and not exists ( -- 就是加一个not
    select * from courses c
    where c.dept = s.dept
    and not exists (
        select * from enrollment e
        join courseclass cc on cc.clsno = e.clsno
        where e.sno = s.sno
        and cc.cno = c.cno
        and e.grade >= 60
    )
)
order by s.dept asc, s.sno asc;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

dept	sno	sname
计算机	211210001	陈圣威
计算机	211210091	郭静
计算机	211210119	王伟
人工智能	211210042	刘菲
人工智能	211210112	李杰
数学	211210155	李程锦
NULL	NULL	NULL

9.

select t.tno, t.tname, t.dept,

min(year(cc.clsyear)) as first_year, -- 天然保留 null 值，表示没有开过课的教师

max(year(cc.clsyear)) as last_year

from teachers t

left join courseclass cc on t.tno = cc.tno -- 如果不用 left join，则无法统计没有开课的教师，也就无法保留空值

group by t.tno

order by t.dept asc, t.tno asc;

```sql
select t.tno, t.tname, t.dept,
    min(year(cc.clsyear)) as first_year, -- 天然保留null值，表示没有开过课的教师
    max(year(cc.clsyear)) as last_year
from teachers t
left join courseclass cc on t.tno = cc.tno -- 如果不用left join，则无法统计没有开课的教师，也就无法保留空值
group by t.tno
order by t.dept asc, t.tno asc;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

tno	tname	dept	first_year	last_year
602	朱荣华	计算机	NULL	NULL
704	张家荣	计算机	NULL	NULL
777	李云龙	计算机	NULL	NULL
811	李晋城	计算机	NULL	NULL
703	陈坤	人工智能	NULL	NULL
802	张无忌	人工智能	NULL	NULL
601	王家庚	数学	NULL	NULL
805	杨玉昆	数学	NULL	NULL

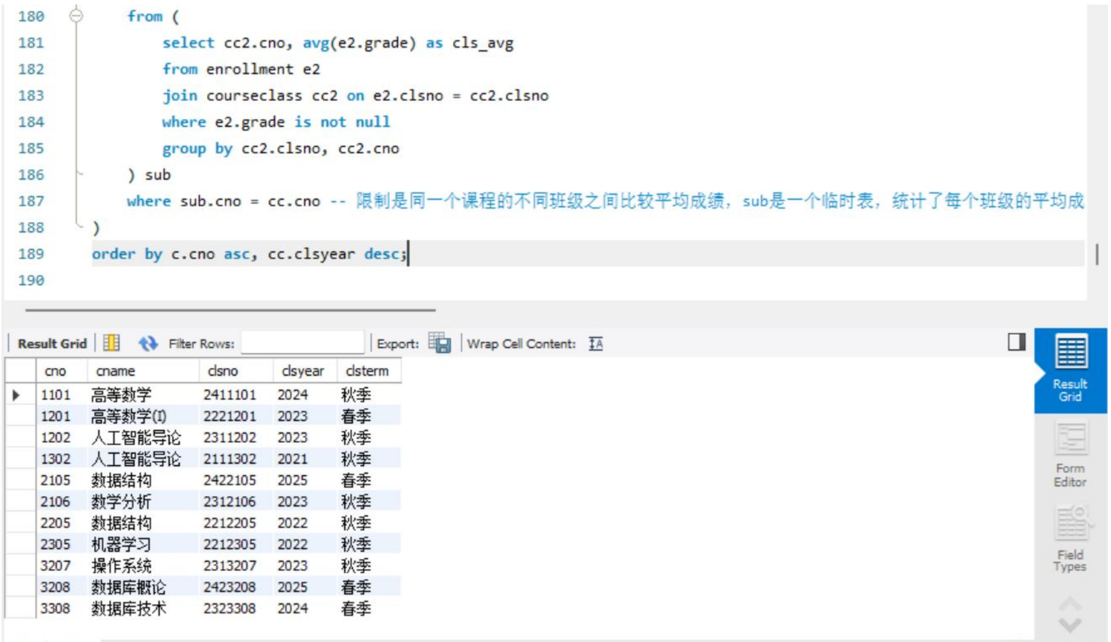
10.

select c.cno, c.cname, cc.clsno, cc.clsyear, cc.clsterm

from courseclass cc

join courses c on cc.cno = c.cno

```
join (
	select clsno, avg(grade) as cls_avg
	from enrollment
	where grade is not null
	group by clsno
) ca on ca.clsno = cc.clsno -- classnumber and avarage grade for each class
where ca.cls_avg = (
	select max(sub.cls_avg)
	from (
		select cc2.cno, avg(e2.grade) as cls_avg
		from enrollment e2
		join courseclass cc2 on e2.clsno = cc2.clsno
		where e2.grade is not null
		group by cc2.clsno, cc2.cno
	) sub
	where sub.cno = cc.cno -- 限制是同一个课程的不同班级之间比较平均成绩，sub 是一个临时表，统计了每个班级的平均成绩，然后在外层查询中找出每个课程的最高平均成绩
)
order by c.cno asc, cc.clsyear desc;
```



第四部分：数据更新

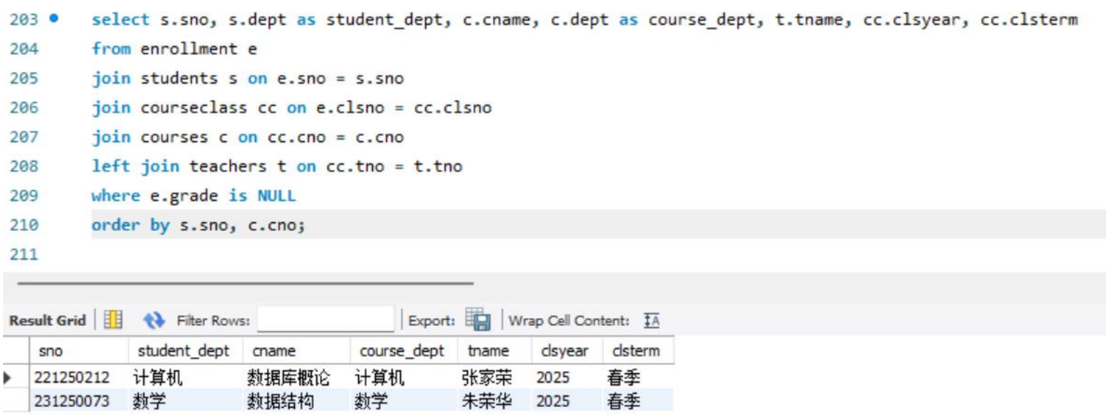
（一）元组插入

```
set autocommit=0;
insert into courseclass values('2513208','3208', '704', '2025', '秋季');
```

insert into enrollment(sno, clsno, grade)
select sno, '2513208', null
from students
where dept = '计算机学院' and year(enrolldate) = 2024;

COMMIT;

select s.sno, s.dept as student_dept, c.cname, c.dept as course_dept, t.tname, cc.clsyear, cc.clsterm
from enrollment e
join students s on e.sno = s.sno
join courseclass cc on e.clsno = cc.clsno
join courses c on cc.cno = c.cno
left join teachers t on cc.tno = t.tno
where e.grade is NULL
order by s.sno, c.cno;



（二）元组修改

set autocommit=0;
update enrollment
set grade = 68
where sno = '211210166' and clsno = '2111202';

select s.sno, s.sname, s.dept
from students s
where year(s.enrolldate) = 2021
    and not exists(
        select * from courses c
        where c.dept = '计算机学院'
        and not exists(

```
                select * from enrollment e
                join courseclass cc on e.clsno = cc.clsno
                where e.sno = s.sno
                and cc.cno = c.cno
                and e.grade >= 60
            )
        )
order by s.sno;

ROLLBACK;
```

```
from students s
where year(s.enrolldate) = 2021
    and not exists(
        select * from courses c
        where c.dept = '计算机学院'
        and not exists(
            select * from enrollment e
            join courseclass cc on e.clsno = cc.clsno
            where e.sno = s.sno
            and cc.cno = c.cno
            and e.grade >= 60
        )
    )
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

sno	sname	dept
211210001	陈圣威	计算机
211210042	刘菲	人工智能
211210043	刘家君	计算机
211210088	吴小芳	数学
211210091	郭静	计算机
211210112	李杰	人工智能
211210119	王伟	计算机
211210155	李程锦	数学
211210166	蒋超	计算机
NULL	NULL	NULL

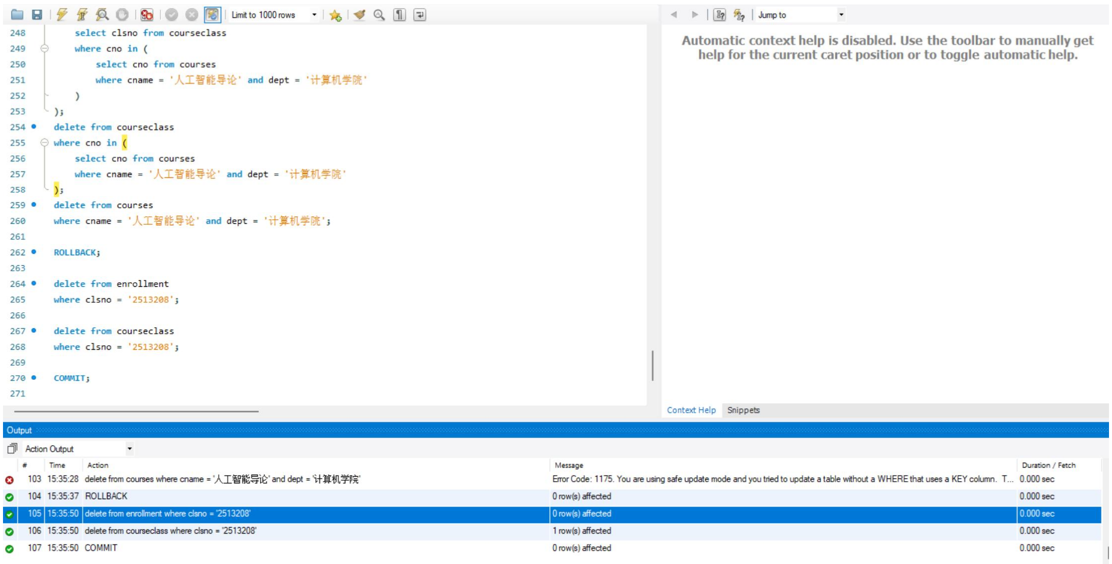
```
update enrollment
join students s on enrollment.sno = s.sno
join courseclass cc on enrollment.clsno = cc.clsno
join courses c on cc.cno = c.cno
set grade = NULL
where c.dept != s.dept;

ROLLBACK;
```

✔	97	15:28:35	ROLLBACK	0 row(s) affected
✔	98	15:28:38	update enrollment join students s on enrollment.sno = s.sno join courseclass cc on enrollment.clsno = cc.clsno join courses c on c...	23 row(s) affected Rows matched: 23 Changed: 23 Warnings: 0
✔	99	15:29:00	ROLLBACK	0 row(s) affected

（三）元组删除

```
set autocommit=0;
delete from enrollment
where clsno in (
        select clsno from courseclass
```

```
    where cno in (
        select cno from courses
        where cname = '人工智能导论' and dept = '计算机学院'
    )
);
delete from courseclass
where cno in (
    select cno from courses
    where cname = '人工智能导论' and dept = '计算机学院'
);
delete from courses
where cname = '人工智能导论' and dept = '计算机学院';

ROLLBACK;

delete from enrollment
where clsno = '2513208';

delete from courseclass
where clsno = '2513208';

COMMIT;
```

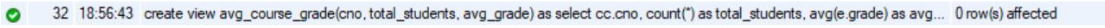


第五部分：视图创建与访问

1.

```
create view avg_course_grade(cno, total_students, avg_grade) as
select cc.cno, count(*) as total_students, avg(e.grade) as avg_grade
```

```
from courseclass cc
left join enrollment e on cc.clsno = e.clsno
where e.grade is not null
group by cc.cno;
```

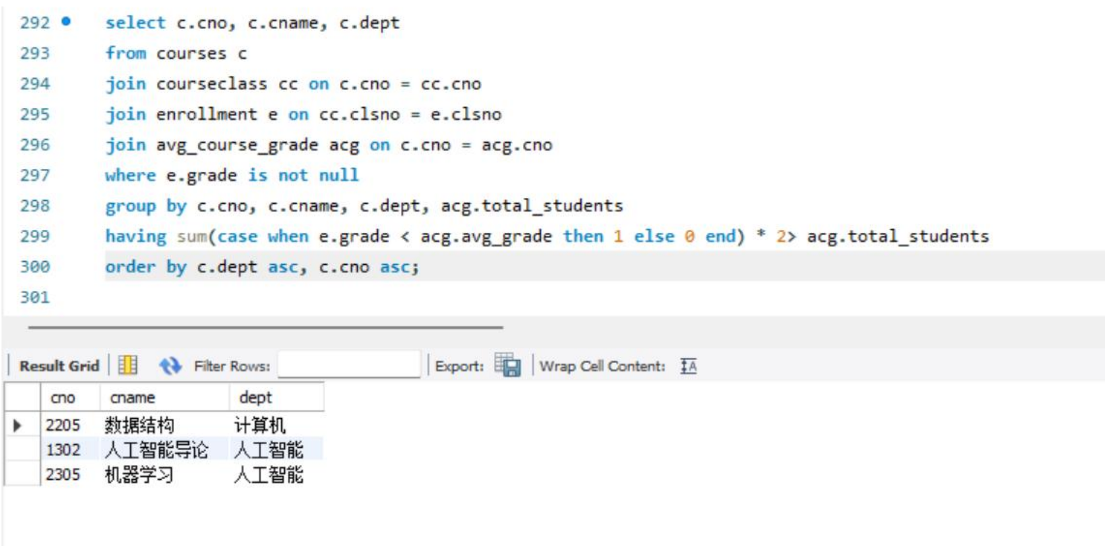


2. （1）

```
create view total_course_students(cno, total_students, num_greater_avg_grade) as
select cc.cno,
       count(*) as total_students,
       sum(case when e.grade > acg.avg_grade then 1 else 0 end) as
num_greater_avg_grade
from courseclass cc
left join enrollment e on cc.clsno = e.clsno
left join avg_course_grade acg on cc.cno = acg.cno
group by cc.cno;
```

（2）

```
select c.cno, c.cname, c.dept
from courses c
join courseclass cc on c.cno = cc.cno
join enrollment e on cc.clsno = e.clsno
join avg_course_grade acg on c.cno = acg.cno
where e.grade is not null
group by c.cno, c.cname, c.dept, acg.total_students
having sum(case when e.grade < acg.avg_grade then 1 else 0 end) * 2> acg.total_students
order by c.dept asc, c.cno asc;
```



3.

（1）

create view failing_enrollments(sno, sname, dept, cno, clsno, grade) as
select s.sno, s.sname, s.dept, c.cno, cc.clsno, e.grade
from enrollment e
join students s on e.sno = s.sno
join courseclass cc on e.clsno = cc.clsno
join courses c on cc.cno = c.cno
where e.grade is not null or e.grade < 60;
with check option;

（2）

select sno, sname, dept, cno, clsno, grade
from failing_enrollments；

```
create view failing_enrollments(sno, sname, dept, cno, clsno, grade) as
select s.sno, s.sname, s.dept, c.cno, cc.clsno, e.grade
from enrollment e
join students s on e.sno = s.sno
join courseclass cc on e.clsno = cc.clsno
join courses c on cc.cno = c.cno
where e.grade is null or e.grade < 60
with check option;

select sno, sname, dept, cno, clsno, grade
from failing_enrollments
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

sno	sname	dept	cno	clsno	grade
211210088	吴小芳	数学	2106	2212106	53
211210166	蒋超	计算机	1202	2111202	55
221250212	陈凯歌	计算机	3208	2423208	NULL
231250030	王泳桦	计算机	2205	2412205	57
231250073	王东阳	数学	2105	2422105	NULL
241210032	陈玲	人工智能	1201	2421201	41

（3）

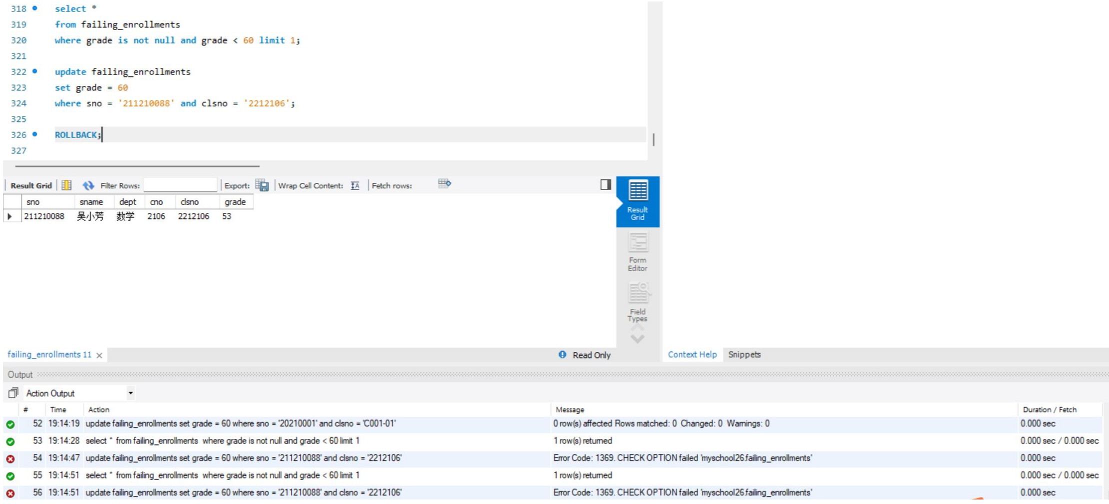
set autocommit=0;

（4）（5）

select *
from failing_enrollments
where grade is not null and grade < 60 limit 1;

update failing_enrollments
set grade = 60
where sno = '211210088' and clsno = '2212106';

ROLLBACK;



（6）（7）（8）

update enrollment e

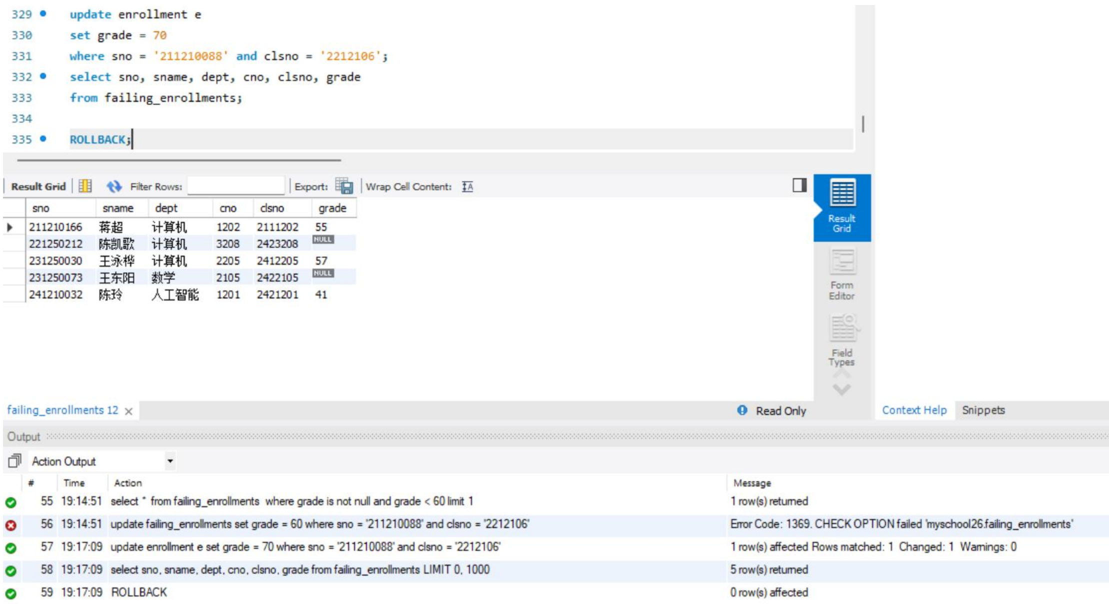
set grade = 70

where sno = '211210088' and clsno = '2212106';

select sno, sname, dept, cno, clsno, grade

from failing_enrollments;

ROLLBACK;



（9）

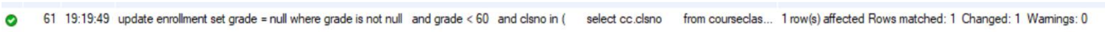
update enrollment e

set grade = 50

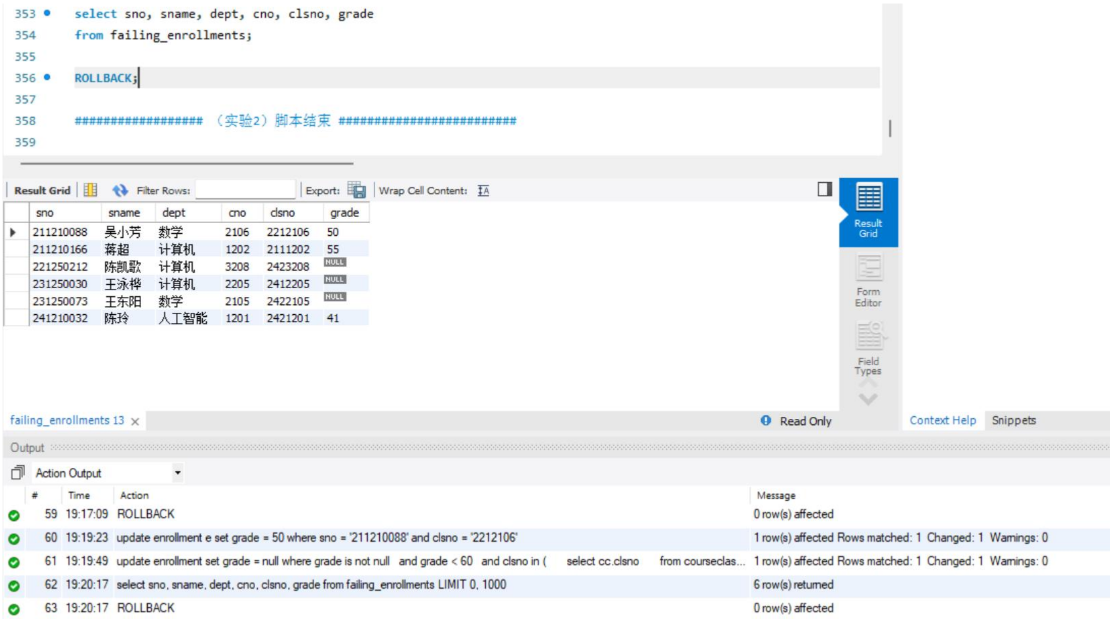
where sno = '211210088' and clsno = '2212106';

（10）

```
update enrollment
set grade = null
where grade is not null
   and grade < 60
   and clsno in (
          select cc.clsno
          from courseclass cc
          join courses c on cc.cno = c.cno
          where c.cname = '数据结构'
);
```

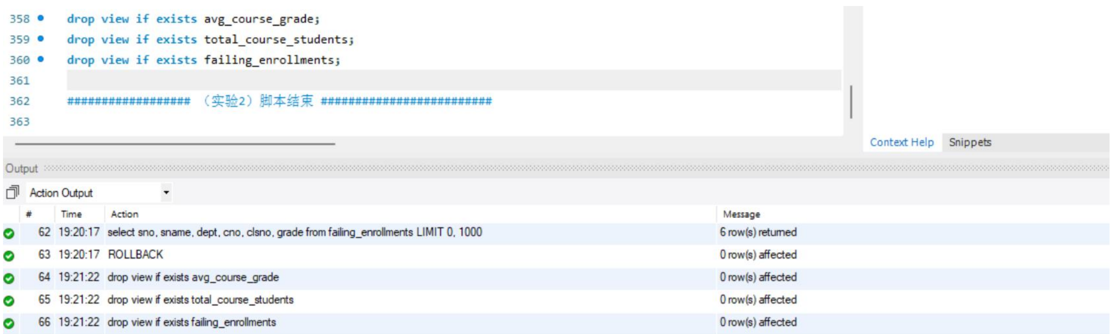


（11）（12）



4.

```
drop view if exists avg_course_grade;
drop view if exists total_course_students;
drop view if exists failing_enrollments;
```



实验中遇到的困难及解决办法

[详细说明你认为本次实验中比较困难的地方，也可以对实验设计提出建议]

- 一些不熟悉的函数需要搜索用法，比如字符串函数 left
- 没有弄清楚 where 和 having 的用法
- 对视图的操作不熟练
- case when 子句不熟练

参考文献及致谢

[如果你参考了任何书籍、网页，或与他人进行了讨论，请在此注明]

https://claude.ai/chat 询问字符串函数 left 的用法，以及'_'和'%'的区别